

Estimating Magnitude Completeness in Earthquake Catalogs: A Comparative Study of Catalog-Based Methods

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Key Points:

- Evaluated nine catalog-based methods for estimating the completeness magnitude (M_c) in both simulated and observed earthquake catalogs
- Introduced BSReLU and AEReLU as augmented Gutenberg-Richter models, introducing a novel probabilistic framework for modeling the magnitude M_c of completeness
- Demonstrated that AEReLU is the most reliable method even under the assumption of a sharp cut-off M_c , despite being designed for a gradual probabilistic transition from incomplete to complete detection

Supporting Information:

Supporting Information may be found in the online version of this article.

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Abstract Without careful attention to the earthquake catalog completeness, claims of novel discoveries or forecasting skills lack credibility. Estimating the completeness magnitude (M_c) is therefore a critical step in seismological analysis. Among the available techniques, catalog-based methods are the most accessible and widely adopted, and they also often form the basis for more sophisticated techniques. However, current frameworks for evaluating methods do not provide a standardized strategy for generating synthetic catalogs that are independent of any specific M_c estimation technique. An effective evaluation framework should also allow for the simulation of data sets with spatially and temporally varying M_c values. In this study, we introduce a robust evaluation framework specifically designed to benchmark catalog-based M_c estimation methods, including six established and three newly proposed methods, under realistic and controlled simulation conditions. The proposed methods are evaluated on both simulated data sets, with homogeneous and heterogeneous M_c distributions, and on real-world earthquake catalogs from China, California, and New Zealand. In the synthetic tests, The traditional method MBS-WW and two new methods, BSReLU and especially AEReLU, consistently deliver reliable M_c estimates, provided the data are sufficiently dense and well-resolved. Among them, AEReLU demonstrates superior accuracy and adaptability, particularly under challenging conditions. When applied to six empirical catalogs, AEReLU produces the most robust and physically meaningful M_c estimates. Its predicted M_c values closely align with known regional completeness thresholds (e.g., $M_c \approx 1.8$) and remain stable across diverse tectonic environments. Moreover, AEReLU yields consistent b -values and reveals spatial variations in the asymmetry parameter β , which captures differences in how detection completeness converges below and above M_c , reflecting region-specific patterns of seismicity and catalog incompleteness. Unlike traditional methods that impose a sharp M_c cut-off, BSReLU and AEReLU adopt a probabilistic framework that models the smooth transition in detection likelihood from zero to one as magnitude increases. This formulation better reflects the gradual shift from incomplete to complete reporting, effectively overcoming the limitations of step-function-based models. By rigorously comparing these methods, the present study not only identifies the contexts in which each is most suitable but also enhances our understanding of seismicity, earthquake forecasting, and hazard assessment, underscoring the critical role of data completeness in all such analyses.

Plain Language Summary In earthquake studies, determining the smallest magnitude at which all earthquakes are detected, known as the completeness magnitude (M_c), is essential for accurate analysis. Various methods exist to estimate M_c , but it is crucial to identify which are most reliable under different conditions. This study evaluates nine catalog-based methods, including three newly developed approaches, by testing them on both simulated and real-world earthquake data from regions like China, California, and New Zealand. Among these, AEReLU allowing for an asymmetric transition from incompleteness to completeness, demonstrates superior performance. We introduce AEReLU and BSReLU as two novel approaches that model the transition from complete to incomplete earthquake data as a gradual probabilistic process rather than a sharp transition. This framework provides a more realistic representation of detection capabilities in earthquake catalogs. By gaining deeper insight into the strengths and limitations of both traditional and newly proposed models, we can improve the accuracy of earthquake forecasting and enhance the reliability of seismic hazard assessments.

1. Introduction

The completeness magnitude (M_c) is the threshold magnitude above which all earthquakes are reliably recorded within a specified spatial-temporal window (Mignan & Woessner, 2012; Rydelek & Sacks, 1989; Woessner &

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Wiemer, 2005). It represents the smallest magnitude at which the catalog is deemed “complete,” meaning all earthquakes of this magnitude or larger are recorded with near certainty (probability close to 1). These events adhere to the Gutenberg-Richter (GR) law, which describes the dependence of the complementary cumulative frequency-magnitude distribution (CCFMD) as being a decaying exponential function of the magnitude:

$$\text{CCFMD}(M) = 10^{a-bM}, \quad M \geq M_c \quad (1)$$

Here, a and b are constants, with b , the slope of the GR law, typically close to 1. Below the magnitude of completeness, smaller earthquakes are recorded with a probability smaller than 1, depending on the detection capabilities of the seismic monitoring network.

Accurately estimating M_c is crucial for reliable seismic hazard assessments, earthquake forecasting, and understanding the underlying processes that govern seismicity. Indeed, many purported discoveries of new patterns, structures or laws in earthquake forecasting have later been attributed to biases caused by incomplete earthquake catalogs (e.g., Li, Sornette, Wu, et al., 2025; Seif et al., 2017; Zhuang et al., 2017). Without rigorous attention to the completeness of these catalogs, no claim of new discoveries or forecasting skills—whether derived from back tests, statistical analyses, or advanced modeling—can be deemed credible. The integrity of earthquake catalog data is fundamental, as any overlooked gaps or inconsistencies may lead to false positives, misinterpretations, and unreliable predictions. Only by addressing these issues with the utmost diligence can meaningful advancements in seismic research and forecasting be achieved. Therefore, an accurate estimation of M_c has become one of the essential first steps in nearly all robust studies of earthquake statistical physics (Mignan & Woessner, 2012).

Many methods have been developed to assess M_c , which can be broadly grouped into three categories:

- (a) Catalog-based methods rely solely on earthquake catalogs as input data. These methods are typically easier to implement and use data that are more readily available. Common catalog-based methods include the maximum curvature (MAXC; Wiemer & Wyss, 2000; Wyss et al., 1999) method, goodness-of-fit (GFT) tests (Wiemer & Wyss, 2000), the b -value stability method (MBS; Cao & Gao, 2002; Woessner & Wiemer, 2005), the entire magnitude range (EMR) method (Woessner & Wiemer, 2005), median-based analysis of the segment slope (MBASS; Amorèse, 2007), and day-to-night noise modulation (Rydelek & Sacks, 1989). Among them, MAXC and MBASS are non-parametric techniques based on the frequency-magnitude distribution (FMD). In recent years, Herrmann and Marzocchi (2021) proposed a method for analyzing M_c by testing the exponentiality of FMD, using the Lilliefors test (Lilliefors, 1969). Building on this approach, Taroni (2023) explored whether the method for estimating M_c by testing the exponentiality could be applied to earthquake catalogs composed of mixed exponential distributions, such as those containing events with varying b -values. However, much like the Kolmogorov-Smirnov statistic-based approach proposed by Clauset et al. (2009) for determining the minimum cut-off sample size necessary to preserve a power law distribution, these test-based methods are computationally intensive. This is due to the requirement of generating a large number of synthetic samples to compute p -values. Consequently, their computational efficiency is notably low.
- (b) Network-based methods, such as the Probabilistic Magnitude of Completeness (PMC; Schorlemmer & Woessner, 2008) and others discussed in Section 5.2 of Mignan and Woessner (2012), approach M_c from a network detection perspective. The primary goal of these methods is not to define the completeness level of the catalog, but rather to estimate the probability that an earthquake can be detected. These methods require input data such as seismic waveforms, phase-pick data, station metadata, and attenuation relations used for magnitude determination. However, they face challenges related to both data availability and the complexity of computation and result interpretation.
- (c) Hybrid methods combine one or more catalog-based methods with other techniques. For example, Mignan and Woessner (2012) proposed a combined approach to estimating M_c : when the GFT test does not reach the 95% confidence level, the fitting level is lowered to 90%. If M_c cannot be estimated with this adjustment, the MAXC method is used. This strategy is implemented in the ZMAP software (Wiemer, 2001). Another example is the Bayesian Magnitude of Completeness (BMC) proposed by Mignan et al. (2011), which estimates posterior M_c by integrating M_c estimates obtained using one of the catalog-based methods (e.g., MAXC in Feng et al., 2022; Mignan et al., 2011; MBASS in Li et al., 2023) with prior Bayesian information, weighted by their respective standard deviations. This prior information is derived from the relationship between the density of the seismic network and M_c estimates. More recently, Mignan (2019) introduced a

mixture of asymmetric Laplace distributions to describe magnitude data that range from uniform to highly heterogeneous. Hence, hybrid methods strongly rely on the reliability and robustness of the catalog-based methods used as their foundational algorithms.

In summary, catalog-based methods are straightforward to implement using readily available data and often serve as the foundation for more advanced M_c analysis techniques. Therefore, it is crucial to understand which of these methods are more reliable, and under what circumstances. Existing comparisons primarily assess the accuracy and stability of these methods using observed seismicity with unknown M_c or simulated seismicity with known M_c (Herrmann & Marzocchi, 2021; Lombardi, 2021; Mignan, 2019; Mignan & Woessner, 2012; Pavlenko & Zavyalov, 2022; Taroni, 2023; Woessner & Wiemer, 2005; Zhou et al., 2018). For observed data, the consistency of M_c estimates across different methods applied to the same data is also tested (Herrmann & Marzocchi, 2021; Kagan, 2003; Mignan, 2019; Mignan & Woessner, 2012). However, most of these comparison strategies overlook the critical fact that M_c can vary over time and space due to heterogeneities in station distribution, site conditions, and sensor sensitivity (e.g., Feng et al., 2022; Godano et al., 2021; Li et al., 2023; Mignan & Woessner, 2012; Mignan et al., 2011). Furthermore, existing models commonly used to synthesize incomplete data assume that FMD is a mixture of Normal (Gaussian) and power law (Gutenberg-Richter) components, with the magnitude of transition being identified as M_c (Ogata & Katsura, 1993, 2006; Woessner & Wiemer, 2005). However, this synthesis model represents a specification of the EMR method (Section 2), and thus, a mis-specification of other methods. Additionally, the asymmetry in incomplete segments often leads to discontinuities at the transformed magnitude M_c between the Gaussian and power-law components. Other approaches have been also proposed to model the incomplete portion of the FMD, such as using another exponential distribution (ANGular model; Mignan, 2012) or a quadratic polynomial distribution (POL model; García-Hernández et al., 2019). Although both differ markedly from the empirical FMD curves we observe (see Figure S2 in Supporting Information S1), it is worth noting that the ANGular model, originally formulated for a catalog with uniform completeness, can reproduce observed FMDs when M_c varies spatially or temporally, provided the catalog is treated as the superposition of many local exponential tails, each anchored to its own M_c (Mignan, 2012, 2019; Mignan & Chen, 2016). Therefore, there is a need for a unified simulation strategy for generating data that is not tied to any specific M_c estimation method, and this strategy should also simulate data with a non-uniform M_c distribution.

The purpose of the present paper is to assess the accuracy and reliability of commonly used catalog-based methods for estimating M_c using both simulated and observed data that reflect the heterogeneous temporal and spatial distribution of M_c . We introduce three new catalog-based methods for estimating M_c , drawing on our expertise in both statistical testing and computer science, and compare them with other existing methods. Additionally, we propose a novel seismicity pruning framework that not only simulates incomplete seismicity through a pruning function or the ANGular model, but also incorporates the modeling of the spatiotemporal distribution of this incompleteness. By examining the limitations of these foundational catalog-based methods in estimating M_c , we aim to refine our understanding of their appropriate applications, ultimately improving insights into seismicity patterns, earthquake forecasting, and hazard mitigation by emphasizing the importance of data-quality considerations.

2. Catalog-Based Methods for Estimating M_c

In this section, we appraise nine catalog-based methods, including three new methods developed by drawing on our expertise in both statistical testing and computer science. Methods (1) through (6) are previously established and widely used techniques for estimating the completeness magnitude M_c , while methods (7) through (9) are novel contributions proposed in the present study. The present paper does not discuss the day-to-night noise modulation method proposed by Rydelek and Sacks (1989), although it is also a catalog-based method. This is because the method requires the prior removal of other non-random features from the earthquake catalog, such as earthquake clusters. These requirements impose significant limitations on its applicability (Wiemer & Wyss, 2003).

2.1. Established Methods

1. MAXC

The maximum curvature (MAXC) method, proposed by Wyss et al. (1999), determines M_c by identifying the magnitude corresponding to the point of maximum curvature, calculated as the maximum value of the first derivative of the FMD, which is itself the first derivative of the CCFMD, making it a fast and straightforward

approach. Generally, the M_c obtained by MAXC, which is typically used as the initial condition in many of the algorithms introduced later, corresponds to the magnitude with the highest event frequency (the most probable value or mode) in the FMD.

2. GFT-95%

The goodness-of-fit tests (GFT), proposed by Wiemer and Wyss (2000), estimate M_c by assessing the goodness of fit between the observed and predicted cumulative frequency-magnitude distributions, namely $\text{CFMD}^{\text{obs}}(M)$ and $\text{CFMD}^{\text{pred}}(M)$, by systematically scanning across magnitudes. These magnitudes are used as the cut-off magnitude, M_{co} , to calculate $\text{GFT}(M_{\text{co}})$ as follows:

$$\text{GFT}(M_{\text{co}}) = 100 - \left[\frac{\sum_{M=M_{\text{co}}}^{M_{\text{max}}} |\text{CFMD}^{\text{obs}}(M) - \text{CFMD}^{\text{pred}}(M)|}{\sum_{M=M_{\text{co}}}^{M_{\text{max}}} \text{CFMD}^{\text{obs}}(M)} 100 \right] \quad (2)$$

Here, M_{max} is the maximum magnitude of the catalog events. The GFT-95% defines M_c as the M_{co} at which $\text{GFT}(M_{\text{co}})$ first falls below the 95% fitting level (or confidence level) as magnitudes are scanned by decreasing values.

3. MBS-CG

Cao and Gao (2002) proposed a method for estimating M_c by analyzing the stability of the b -value as a function of M_{co} . This method, referred to as the method of b -value stability (MBS) by Woessner and Wiemer (2005), is investigated in the present paper. To distinguish it from an improved version of MBS introduced later, we refer to it here as MBS-CG. The theoretical foundation of this method is that the estimated b -value behaves as follows: it increases with M_{co} when $M_{\text{co}} < M_c$, stabilizes when $M_{\text{co}} \geq M_c$, and fluctuates significantly when $M_{\text{co}} \gg M_c$. Specifically, for $M_{\text{co}} < M_c$, the FMD deviates from the GR law, falling below the GR law extrapolation at smaller magnitudes, leading to inaccurate b -value estimates. As M_{co} approaches M_c , the estimated b -value gradually converges to its true value and remains stable when $M_{\text{co}} \geq M_c$. However, as M_{co} increases significantly beyond M_c , the sample size decreases, particularly in the right tail of the distribution, causing strong fluctuations in the b -value. Cao and Gao (2002) defined M_c as the magnitude at which the change in b -value (Δb) between two consecutive M_{co} values first becomes less than 0.03 as M_{co} is increased.

4. MBS-WW

Woessner and Wiemer (2005) identified that the instability of the MBS-CG method was due to the significant variability in the event frequency in each magnitude bin. To enhance the reliability of the b -value stability estimation method, they incorporated the more objective b -value uncertainty index, δb , proposed by Shi and Bolt (1982), as a criterion:

$$\delta b = 2.3b^2 \sqrt{\frac{\sum_{i=1}^N (M_i - M_{\text{ave}})^2}{N_{\text{evt}}(N_{\text{evt}} - 1)}} \quad (3)$$

Here, M_{ave} is the average magnitude of the catalog events, N_{evt} is the number of events. M_c is then defined as the first M_{co} at which $\Delta b(M_{\text{co}}) = |b(M_{\text{co}}) - b_{\text{ave}}| \leq \delta b$ as M_{co} is increased. The arithmetic mean of the b -values, b_{ave} , is calculated by averaging all b -values within a specified magnitude range (ΔM) according to:

$$b_{\text{ave}} = \frac{\sum_{M=M_{\text{co}}}^{M_{\text{co}}+\Delta M} b(M)dM}{\Delta M} \quad (4)$$

Typically, the magnitude bin dM is set to 0.1 and the magnitude range ΔM is set to 0.5. To distinguish this improved method introduced by Woessner and Wiemer (2005) from the original MBS-CG, we refer to it as MBS-WW.

5. EMR

The entire magnitude range (EMR) method, proposed by Woessner and Wiemer (2005), estimates M_c by simultaneously considering both the complete and incomplete parts of the FMD, aiming to replicate the properties of the entire distribution. For the complete part ($M \geq M_c$), the GR law distribution of Equation 1 is assumed. For the incomplete part ($M < M_c$), a Normal cumulative distribution function $q(M|\mu, \sigma)$ proposed by Ringdal (1975) and Ogata and Katsura (1993, 2006) is employed to model the detection probability of the seismic network for earthquakes of small magnitudes. The function $q(M|\mu, \sigma)$ is expressed as:

$$q(M|\mu, \sigma) = \begin{cases} \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{M_c} \exp\left[-\frac{(M-\mu)^2}{2\sigma^2}\right] dM, & M < M_c \\ 1, & M \geq M_c \end{cases} \quad (5)$$

Here, μ represents the magnitude at which 50% of earthquakes are detected by the seismic network, while σ represents the width of the identified incomplete portion of the magnitude range (i.e., the standard deviation of the normal cumulative distribution function; Ogata & Katsura, 1993, 2006). A larger σ implies a slower decline in the monitoring capacity of the seismic network as the magnitude decreases. For $M \geq M_c$, the detection probability is assumed to be 1, aligning with the CCFMD described by Equation 1. This dual approach allows the EMR method to effectively characterize both the complete and incomplete portions, providing a comprehensive estimate of M_c .

6. MBASS

The median-based analysis of the segment slope (MBASS) method, proposed by Amorè (2007), employs a non-parametric statistical approach to detect change points in the FMD. This method utilizes the multiple change-point procedure developed by Lanzante (1996), an iterative technique for identifying multiple change points in time series data. Lanzante's method identifies points at which the median of a time series changes (Siegel & Castellan, 1988). Amorè (2007) applied this approach to the segmented slopes of FMDs. The process begins by dividing the set of magnitudes into n_{seg} segments, each defined by magnitude boundaries M_i ($i = 1, \dots, n_{\text{seg}}$). The slope $s(M_i)$ for the i -th segment is then calculated based on the FMD within these boundaries:

$$s(M_i) = \frac{\log[\text{FMD}(M_{i-1})] - \log[\text{FMD}(M_i)]}{M_{i-1} - M_i} \quad (6)$$

Next, the rank sum SR_i for the i -th segment is calculated as the sum from 1 up to its position in the ordered set $s\{M_q\}$, where $q = 1, \dots, n_{\text{seg}}$. This is followed by adjusting the rank sum to obtain SA_i :

$$SA_i = \left| \left(2 \sum_{j=1}^i SR_j \right) - i(n_{\text{seg}} + 1) \right| \quad (7)$$

The maximum value of SA_i is then identified, and the corresponding point $n_{\text{max}}(SA)$ is used to divide $s(M)$ into two groups. A null hypothesis is formulated that assumes no discontinuity in $s(M)$ at $n_{\text{max}}(SA)$. The Wilcoxon rank-sum test (Wilcoxon, 1945) is applied to determine whether the null hypothesis can be rejected. The test is performed at a significance level of 0.05. If a p -value < 0.05 is obtained, the magnitude associated with the minimum p -value is identified as the estimated M_c . The MBASS technique can identify multiple discontinuities in the FMD. The primary discontinuity typically corresponds to M_c , while other discontinuities may indicate high-magnitude breakpoints (Wesnousky, 1994).

2.2. Newly Developed Methods

7. KST-95%

The Kolmogorov-Smirnov statistic is commonly employed to measure the degree of divergence between a sample data set and its corresponding theoretical distribution (Arshad et al., 2002; Baumgartner & Kolassa, 2021; Clauset et al., 2009; Corral et al., 2011; Gibbons & Chakraborty, 2010; Lombardi, 2021; Molnár & Szokol, 2014). In the present paper, we modify the numerator of the goodness-of-fit tests (GFT) in Equation 2 by replacing it with the

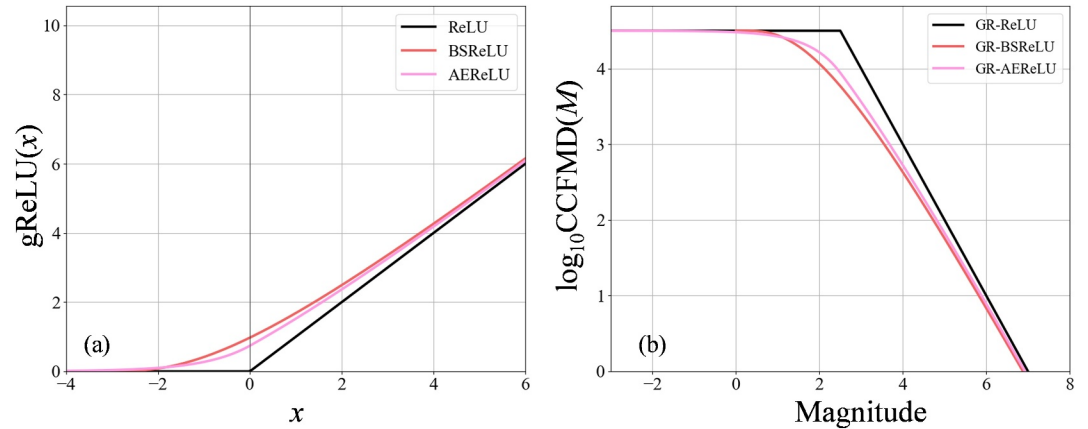


Figure 1. Plots of (a) the two $gReLU(x)$ functions introduced in the text and (b) the corresponding augmented complementary cumulative frequency-magnitude distributions (CCFMDs). For comparison, the figure also includes the Rectified Linear Unit, $ReLU(x)$, depicted with a black solid line, which is defined as $ReLU(x) = 0$ for $x < 0$ and $ReLU(x) = x$ for $x \geq 0$, along with the corresponding CCFMD recovering the pure Gutenberg-Richter law. The relationship between x in (a) and the magnitude M in (b) is given by $x = (M - M_c)/\sigma_{M_c}$. The parameters are set to $a = 7.0$, $b = 1.0$, $M_c = 2.5$, $\sigma_{M_c} = 0.75$, and $\beta = 0.3$, which are values that are informed by empirical data.

Kolmogorov-Smirnov statistic. The denominator is defined as the value of $CFMD^{obs}$ at the magnitude $M_{\max(KS)}$ of maximum distance between $CFMD^{obs}(M)$ and $CFMD^{pred}(M)$. Building upon this, we define a Kolmogorov-Smirnov statistic test (KST) as a function of M_{co} , $KST(M_{co})$, as follows:

$$KST(M_{co}) = 100 - \left[\frac{|CFMD^{obs}(M_{\max(KS)}) - CFMD^{pred}(M_{\max(KS)})|}{CFMD^{obs}(M_{\max(KS)})} \right] 100 \quad (8)$$

KST-95% defines M_c as the M_{co} where $KST(M_{co})$ first drops below the 95% confidence level, when scanning magnitudes by decreasing values.

8. BSReLU

We propose an augmented Gutenberg-Richter model to characterize the CCFMD of earthquake catalogs that include incomplete portions, according to:

$$CCFMD(M) = 10^{a-bG(M)} \quad (9)$$

in which

$$G(M) = M_c + \sigma_{M_c} gReLU(x), \quad x = \frac{M - M_c}{\sigma_{M_c}} \quad (10)$$

where $gReLU(x)$ is a function that converges to 0 for $x \rightarrow -\infty$ and to x for $x \rightarrow +\infty$. A special case is $ReLU(x)$, short for Rectified Linear Unit. It is defined such that $ReLU(x) = 0$ for $x < 0$ and $ReLU(x) = x$ for $x \geq 0$. With the choice $gReLU(x) = ReLU(x)$, $CCFMD(M) = 10^{a-bM}$ for $M \geq M_c$ and $CCFMD(M) = 10^{a-bM_c}$ for $M < M_c$. This corresponds to the exact Gutenberg-Richter law for $M \geq M_c$, with no earthquake considered for $M < M_c$ since $CCFMD(M)$ is constant for $M < M_c$.

Here, we seek a function that smoothly transitions between the Gutenberg-Richter law for $M \geq M_c$ and a gradual depletion of earthquake occurrences for $M < M_c$. In addition to the condition to converge to 0 for $x \rightarrow -\infty$ and to x for $x \rightarrow +\infty$, the smoothing function must ensure that $CCFMD(M)$ is monotonous. Hence, $gReLU(x)$ must be a monotonously increasing function of x . A function that satisfies all these requirement turns out to be a function taken from the formula giving the price of a financial call option according to the Black-Scholes model (Black & Scholes, 1973; Figure 1a):

$$\text{BSReLU}(x) = \left(x + \frac{M_c}{\sigma_{M_c}}\right) \Phi\left(\frac{\ln[(x\sigma_{M_c} + M_c)/M_c]}{\sigma_{M_c}} + \frac{\sigma_{M_c}}{2}\right) - \frac{M_c}{\sigma_{M_c}} \Phi\left(\frac{\ln[(x\sigma_{M_c} + M_c)/M_c]}{\sigma_{M_c}} - \frac{\sigma_{M_c}}{2}\right) \quad (11)$$

In the limit $\sigma_{M_c} \rightarrow 0$, $\text{BSReLU}(x)$ tends to $\text{ReLU}(x)$. Substituting Equation 11 into Equations 9 and 10, the augmented Gutenberg-Richter model incorporates not only the a -value and b -value from the Gutenberg-Richter law of Equation 1 but also introduces two additional parameters, M_c and σ_{M_c} , which quantify the level of completeness of the catalog as a function of magnitude (Figure 1b). In this model, M_c is no longer treated as a hard threshold that separates the complete and incomplete parts of the CCFMD. Instead, it is modeled as the mean position of the transition that has a width given by σ_{M_c} . This replaces the point representation of M_c by a probabilistic description of M_c . In other words, the BSReLU model simultaneously captures the Gutenberg-Richter behavior of the logarithm of CCFMD, which asymptotically aligns with a linear trend for $M \geq M_c$, while also integrating a probabilistic representation of the completeness magnitude. This implies that, even for magnitudes larger than M_c , there is still a small but nonzero probability of missing some large earthquakes. Conversely, this also indicates that, for magnitudes smaller than M_c , the probability of detecting small earthquakes is not zero, but decreases progressively as the magnitude diminishes. By integrating these features, the BSReLU model provides a more physically grounded framework for describing the complexities of real-world earthquake catalogs, particularly those with incomplete entries, offering greater flexibility and comprehensiveness in modeling.

Text S1 of Supporting Information S1 provides two probabilistic interpretations of the BSReLU function. Here we briefly outline their core ideas. In the first probabilistic interpretation, we model detection uncertainty as a multiplicative lognormal perturbation of the true magnitude M and define BSReLU as the expected exceedance of this perturbed magnitude above the completeness threshold M_c ; this yields a smooth, completeness-aware surrogate for M that approaches M when $M \gg M_c$, reverts to M_c when $M \ll M_c$, and transitions between these limits with steepness governed by σ_{M_c} . In the second probabilistic interpretation, let $X(t)$ follow a geometric Brownian motion (GBM) starting at $X(0) = M$ and developing up to time $t = 1$, with drift $(\sigma_{M_c})^2/2$ and volatility σ_{M_c} , so that $\ln X(1) \sim N(\ln M, (\sigma_{M_c})^2)$. We define BSReLU as the expected exceedance of the final position $X(1)$ of the GBM over M_c . This leads to the same closed form expression for the BSReLU function. Both constructions produce the expression in Equation S3 in Supporting Information S1 (and thus recover Equation 11). Under the rescaling $x = (M - M_c)/\sigma_{M_c}$, we obtain the universal curve $\text{BSReLU}_{\text{scaled}}(x)$, providing a coherent probabilistic foundation for the transform.

9. AEReLU

We also introduce another $\text{gReLU}(x)$ as an alternative to the BSReLU function given in Equation 11. This function allows for asymmetric exponential convergences toward 0 for $x < 0$, and toward $x \geq 0$, with the parameter β controlling the degree of asymmetry (hence the name AEReLU; Figure 1a):

$$\text{AEReLU}(x) = \begin{cases} \frac{e^x}{1 + \beta}, & x < 0 \\ x + \frac{e^{-\beta x}}{1 + \beta}, & x \geq 0 \end{cases} \quad (12)$$

This formulation ensures continuity and differentiability at the transition point $x = 0$. Substituting Equation 12 into Equations 9 and 10, the new function retains the same advantages as the BSReLU function (Figure 1b). As with the BSReLU, when expressed in terms of $x = (M - M_c)/\sigma_{M_c}$, the AEReLU function converges to the standard ReLU function $\text{ReLU}(M - M_c)$ in the limit $\sigma_{M_c} \rightarrow 0$, thereby recovering the pure Gutenberg-Richter law for $M \geq M_c$ and a constant for $M < M_c$. The main advantage of AEReLU over BSReLU lies in its flexibility: while BSReLU uses a symmetric Gaussian crossover, AEReLU generalizes this by introducing the asymmetry parameter β , allowing for an asymmetric transition governed by two potentially different rates of exponential decays (rate equal to 1 for $x < 0$ and rate equal to β for $x \geq 0$).

3. Earthquake Catalog

In this section, we introduce the synthetic simulated earthquake catalogs and the observed earthquake catalogs used in our study.

For the synthetic catalogs, we first generate a complete data set of earthquakes with their magnitudes drawn from the GR law, assuming that no event is missed. Next, we model observational incompleteness by discarding events in two distinct ways, rather than relying on a single, assumed-correct mechanism. The first approach applies a linear pruning function, whose physical basis is detailed in Test S2 in Supporting Information S1. The second uses the previously discussed AN (Angular) model, which postulates that events with magnitudes below M_c are detected with an exponentially decreasing probability (Mignan, 2012, 2019; Mignan & Chen, 2016). Since neither removal rule aligns with the completeness models evaluated later, we avoid “model-correct specification,” that is, the practice of generating test data using the same model intended for evaluation. This approach is realistic because, in practice, the true generating model of real catalogs is unknown. Finally, we apply each scheme independently across multiple grids and combine the data to construct synthetic earthquake catalogs with an heterogeneous M_c distribution.

For the observed earthquake catalogs, we select data sets from four regions in Chinese mainland and two additional regions, California and New Zealand, making a total of six regions.

3.1. Simulated Earthquake Catalog

We generate synthetic earthquake catalog magnitude samples using the normalized magnitude probability density function (or normalized frequency-magnitude distribution, FMD) $P(M)$, which is derived from the complementary cumulative density function (CCDF) given by Equation 1. The normalized FMD $P(M)$ is expressed as:

$$P(M) = \frac{b \ln(10) 10^{-bM}}{10^{-bM_{\min}} - 10^{-bM_{\max}}}, \quad M_{\min} \leq M \leq M_{\max} \quad (13)$$

Here, $M_{\min}$ and $M_{\max}$ represent the minimum and maximum magnitudes of earthquakes in the synthetic earthquake catalog. In this study, the parameters for simulating the catalog are set to $b = 1.0$, $M_{\min} = 1.0$, and $M_{\max} = 9.0$. Additionally, the number of simulated earthquake events is set to 10^k (where k takes on values $k = 3, 4, 5, 6$) to represent seismicity counts at varying spatial and temporal scales. Examples of FMD's and CCFMD's of simulated catalogs are shown in Panels (I) of Figures S1(a) through S1(d) in Supporting Information S1.

3.1.1. Modeling Incompleteness With Iterative Pruning

After obtaining the FMD of the synthetic catalog with a magnitude bin size of 0.1, we apply the pruning function described by Equation 14 below to simulate the incomplete observation of earthquake catalogs within a given magnitude range ($m_{\min}, m_{\max}$). Please note that m here refers to the magnitude range used in the pruning function, which is different from M , the earthquake magnitude in the synthetic catalog. To model catalog incompleteness, we construct a linear magnitude pruning function $y(m)$ defined as:

$$y(m) = c_1 m + c_2, \quad m_{\min} \leq m < m_{\max} \quad (14)$$

This function specifies the percentage of missing earthquakes of a given magnitude m ($m_{\min} \leq m \leq m_{\max}$) relative to the complete data set. The parameters c_1 and c_2 are determined by the boundary conditions: at $m = m_{\min}$, $y(m) = 1$, and at $m = m_{\max}$, $y(m) = 0$. In other words, the function assumes that, at $m = m_{\min}$ and lower, 100% of the data is missing, and at $m = m_{\max}$ and higher, 100% of the data is recorded. Between these two extremes, the data loss ratio decreases linearly from $m_{\min}$ to $m_{\max}$. Catalogs of earthquake with a given data loss ratio are generated using random numbers following a Poisson distribution with $y(m)$ as the mean value. In this study, we set $m_{\min} = M_{\min} = 1.0$, while $m_{\max}$ varies from $m_{\min} + 0.1$ to 5.0. Here, $m_{\max}$ represents the true minimum completeness magnitude M_c^{true} after data pruning.

Performing multiple data pruning operations is insightful, as it allows us to simulate a range of incomplete catalogs (whose physical basis is detailed in Test S2 in Supporting Information S1). Panels (II) to (V) of Figures S1(a) through S1(d) in Supporting Information S1 illustrate the results of applying the pruning function iteratively

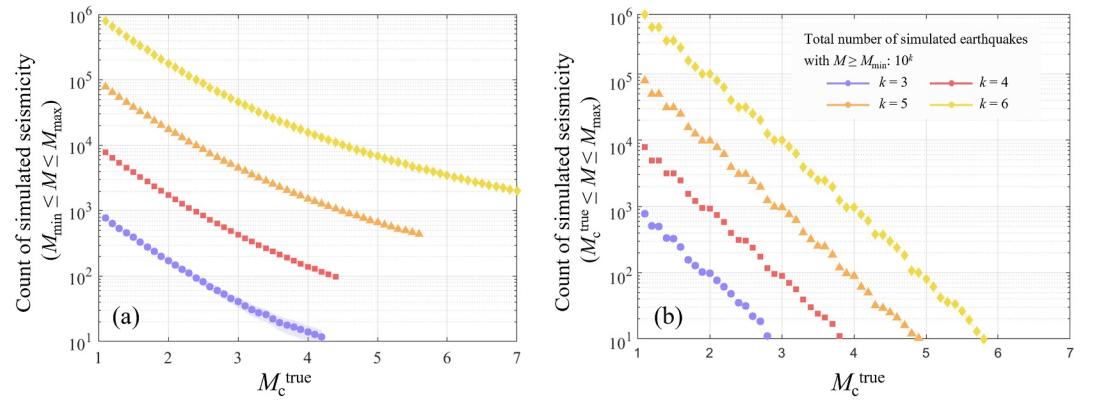


Figure 2. Mean (solid symbols) and standard deviation (shaded area) of earthquake counts from 200 sets of simulated earthquake catalogs after data pruning as a function of the true completeness magnitude M_c^{true} : (a) for magnitudes between the minimum and maximum simulated magnitudes ($M_{\min} = 1.0$ and $M_{\max} = 9.0$), and (b) for magnitudes greater than or equal to M_c^{true} . As an example shown in Figure S1 in Supporting Information S1 for $M_c^{\text{true}} = 2.5$, a linear data pruning function is applied iteratively three times between $M_{\min}$ and M_c^{true} to simulate data loss, modifying the pure Gutenberg-Richter relationship between $M_{\min}$ and $M_{\max}$, and mimicking incomplete seismicity records in real observations. The number of lost seismicity is determined by generating random numbers from a Poisson distribution, using the values calculated by the data pruning function as the mean, which results in the variance (shaded area) observed in the figure. The narrowness of the shaded area is primarily due to the logarithmic scale of the vertical axis. The different colors represent the total number of simulated earthquakes with $M \geq M_{\min}$, given by 10^k with $k = 3, 4, 5, 6$. For details on data pruning, refer to the main text.

from $m_{\min}$ to $m_{\max}$ for 1 to 4 iterations. Figures 2a and 2b, respectively, show the mean and standard deviation of earthquake counts derived from 200 sets of simulated earthquake catalogs as a function of the M_c^{true} for magnitudes between $M_{\min} = 1.0$ and $M_{\max} = 9.0$, and for magnitudes greater than or equal to M_c^{true} (or $m_{\max}$) after 3 iterations of data pruning.

After applying the above data pruning, each data set is characterized by a uniform (or homogeneous) M_c^{true} (or $m_{\max}$). To simulate data sets with spatially and temporally heterogeneous M_c values, the process essentially involves repeating the aforementioned steps for generating M_c^{true} within multiple grids, where each grid is assigned a different M_c^{true} . In this study, for each scenario with total data sizes 10^k ($k = 3, 4, 5, 6$), the data are first generated using Equation 13 without pruning and are then evenly distributed across 2^n grids ($n = 1, 2, \dots, 10$). Within each grid, the data are pruned, and the M_c^{true} (or $m_{\max}$) for each grid is determined as follows: the first and last grids are fixed with $M_c^{\text{true}} = 1.0$ and 5.0 , respectively. For the remaining grids, M_c^{true} is linearly interpolated based on the grid index. For example, in the case of $2^2 = 4$ grids, the first grid has $M_c^{\text{true}} = 1.0$, and the fourth grid has $M_c^{\text{true}} = 5.0$. The second and third grids have $M_c^{\text{true}} = 2.3$ and $M_c^{\text{true}} = 3.7$, respectively, determined by linear interpolation. This construction ensures that the mean M_c^{true} across all grids is 3.0 , and half of the grids have $M_c^{\text{true}} \leq 3.0$, while the other half have $M_c^{\text{true}} > 3.0$. Once M_c^{true} values for all grids are determined, Equation 14 is applied to prune the data in each grid iteratively three times. Finally, the pruned data from all grids are combined to create a data set representing seismicity with heterogeneous M_c values. This approach enables the generation of seismicity that captures spatial and/or temporal variability in M_c .

3.1.2. Modeling Incompleteness With the AN (Angular) Detection Model

As a second avenue for introducing incompleteness, we replace the linear pruning rule of Section 3.1.1 with the AN (Angular) model, which assumes that events below the completeness magnitude M_c are detected with an exponentially decreasing probability (Mignan, 2012, 2019; Mignan & Chen, 2016). The detection probability in each grid is therefore

$$p(m) = \exp[-\kappa(m_{\max} - m)], \quad m_{\min} \leq m < m_{\max} \quad (15)$$

where we adopt $\kappa = 4 \log 10$ following the original studies (Mignan, 2012, 2019; Mignan & Chen, 2016). Here, $m_{\max}$ also represents the true minimum completeness magnitude M_c^{true} . For $m \geq M_c^{\text{true}}$ detection is taken as certain, $p(m) = 1$. While the angular formulation was first proposed for a catalog with constant completeness, the

superposition of many such exponential tails, each anchored to a different local M_c , has been shown to reproduce observed frequency-magnitude distributions (FMDs) when M_c varies in space or time (Mignan, 2012, 2019; Mignan & Chen, 2016).

To generate heterogeneous M_c catalogs with this model, we follow exactly the grid procedure of Section 3.1.1. A fully complete synthetic set of magnitudes is first drawn from Equation 13 and evenly distributed across 2^n grids ($n = 1, 2, \dots, 10$). We consider four different catalog sizes 10^k with $k = 3, 4, 5, 6$. The first and last grids are fixed at $M_c^{\text{true}} = 1.0$ and $M_c^{\text{true}} = 5.0$; intermediate grids receive linearly interpolated M_c^{true} values so that the overall mean equals 3.0. Within every grid, Equation 15 is applied once to stochastically remove events for a given magnitude m ($m_{\min} \leq m \leq m_{\max}$), yielding sets of incomplete catalogs that share the same GR parameters but differ in their spatial (or temporal) completeness pattern. Finally, the thinned grids are merged to form a single catalog whose M_c^{true} varies realistically from one grid to the next.

Each grid may be interpreted as either a spatial cell or a temporal window in which the M_c heterogeneity is minimized. Mixing these grids, whether in space, in time, or in both, creates the large-scale M_c heterogeneity against which the estimation algorithms are tested; the algorithms themselves are insensitive to whether the underlying variation is spatial, temporal, or a combination of the two.

3.2. Observed Earthquake Catalog

We select earthquake catalogs from four regions in the Chinese mainland, as well as California and New Zealand. For the Chinese mainland, we consider the Beijing-Tianjin-Hebei region, the southeastern coastal region, the Sichuan-Yunnan region, and northern Xinjiang. The spatial boundaries of these regions are delineated using the $M_c \approx 1.8$ contour lines derived from the posterior M_c map generated by the BMC method in Li et al. (2023). The data for these regions were sourced from the China Earthquake Networks Center (CENC) and span from 1 January 2009 to 18 November 2024. The number of earthquake events in these catalogs is as follows: 66,380 in the Beijing-Tianjin-Hebei region, 31,421 in the southeastern coastal region, 602,858 in the Sichuan-Yunnan region, and 60,830 in northern Xinjiang. Among these events, the counts of earthquakes with $M \geq 3.0$ are 362, 109, 5,641, and 998, respectively. The California catalog was obtained from the Advanced National Seismic System (ANSS) Comprehensive Earthquake Catalog (ComCat; U.S. Geological Survey, 2017) and spans from 1 January 2009 to 2 November 2023. The spatial boundary is delineated using the $M_c \approx 1.8$ contour lines derived from the posterior M_c map generated by the BMC method in Tormann et al. (2014). This catalog contains 270,458 earthquake events, of which 772 have $M \geq 3$. For New Zealand, the catalog was sourced from the GeoNet Earthquake Catalog (GNS Science, 2022) and spans from 1 January 2009 to 12 December 2023. The spatial boundary is delineated using the station distribution map in Petersen et al. (2011), focusing on areas with dense and evenly distributed seismic stations. This catalog includes 210,549 earthquake events, with 8,450 having $M \geq 3$. For all six observed earthquake catalogs, the maximum focal depth is set to 100 km. Figure S2 in Supporting Information S1 shows the complementary cumulative and density distribution functions of their magnitudes for the observed seismicity in the six study regions.

4. Results

In this section, we present the performance of nine catalog-based methods using simulated catalogs with both homogeneous and heterogeneous M_c distributions, as well as empirical catalogs.

4.1. Estimating M_c for Simulated Seismicity With Homogeneous M_c

We focus on the results obtained after three pruning iterations; outcomes for one, two, and four iterations are presented in Supporting Information S1. This emphasis reflects our hypothesis, illustrated in Text S2 in Supporting Information S1, that seismic station density, instrument quality, and environmental noise act as the three principal “pruning filters” in real settings. Consistently, we observe that catalogs pruned roughly three times produce CCFMD and FMD curves that match the observed seismicity (see Panels IV of Figures S1a–S1d in Supporting Information S1 compared with Figure S2 in Supporting Information S1). Figures 3 and 4 shows the results of estimating the completeness magnitude M_c^{pred} using nine catalog-based methods as a function of the true completeness magnitude M_c^{true} for the simulated catalogs with homogeneous M_c , after being iteratively pruned three times (Section 3.1.1) and using the AN (Angular) detection model (Section 3.1.2). The solid symbols and shaded areas represent the mean and uncertainties of M_c^{pred} derived from 200 sets of simulated catalogs. We

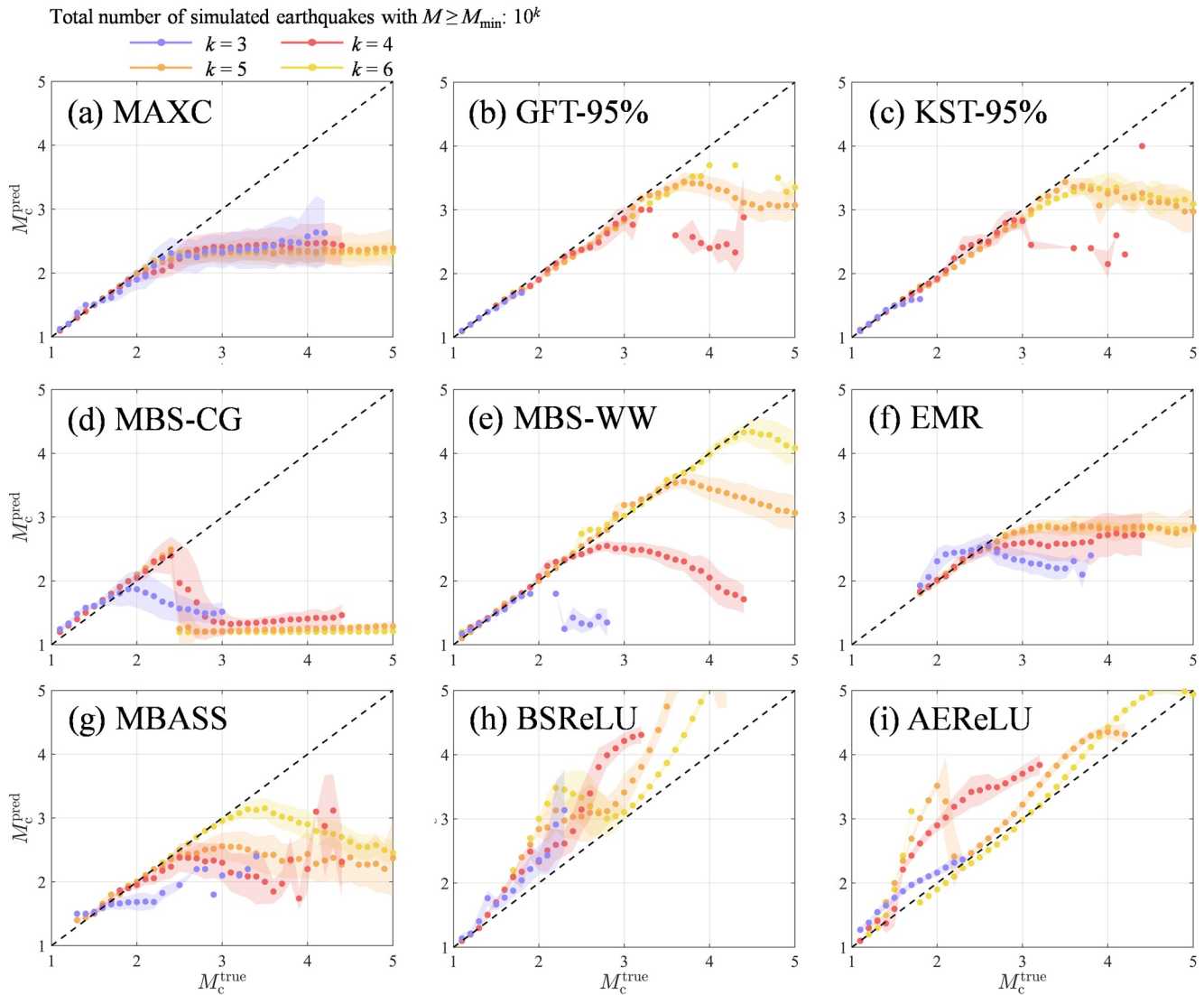


Figure 3. Results of the estimation of the completeness magnitude M_c^{pred} as a function of the true completeness magnitude M_c^{true} for the simulated data sets with homogeneous M_c , after being iteratively pruned three times (Section 3.1.1), as shown in Figure 2 and Panels (IV) of Figure S1 in Supporting Information S1. Results are obtained using the following methods: (a) maximum curvature (MAXC) method, (b) goodness-of-fit test at 95% confidence level (GFT-95%), (c) Kolmogorov-Smirnov statistic test at 95% confidence level (KST-95%), (d) b -value stability method by Cao and Gao (2002) (MBS-CG), (e) b -value stability method by Woessner and Wiemer (2005) (MBS-WW), (f) entire magnitude range (EMR) method, (g) median-based analysis of the segment slope (MBASS), (h) Rectified Linear Unit with Black-Scholes model (BSReLU), and (i) Rectified Linear Unit with asymmetric exponential model (AEReLU). The shaded areas represent the uncertainties in the estimated completeness magnitude M_c^{pred} . In panels (h) BSReLU and (i) AEReLU, M_c^{pred} is defined as the magnitude corresponding to a completeness probability of approximately 84% (see main text for details). The different colors correspond to the total number of simulated earthquakes with $M \geq M_{\min} = 1$, with 10^k for $k = 3, 4, 5, 6$, same as in Figure 2.

define the ratio between Equation 9, which characterizes the observed distribution, and Equation 1, which represents the ideal complete observation, as the completeness probability. This probability can be computed using the fitted parameters. In the BSReLU and AEReLU models, M_c^{pred} is defined as the magnitude corresponding to a completeness probability of approximately 84%, which corresponds to the confidence level of one standard deviation above the mean in a normal distribution. Figures S3–S5 in Supporting Information S1 show the results of M_c^{pred} as a function of M_c^{true} for the simulated catalogs with homogeneous M_c , after being iteratively pruned once, twice, and four times, respectively.

In Figure 4, eight of the nine methods reproduce M_c^{true} very closely; only MBS-CG shows a systematic bias, while BSReLU exhibits a slight systematic bias (0.1–0.2). Moreover, aside from MBS-WW and AEReLU, which retain

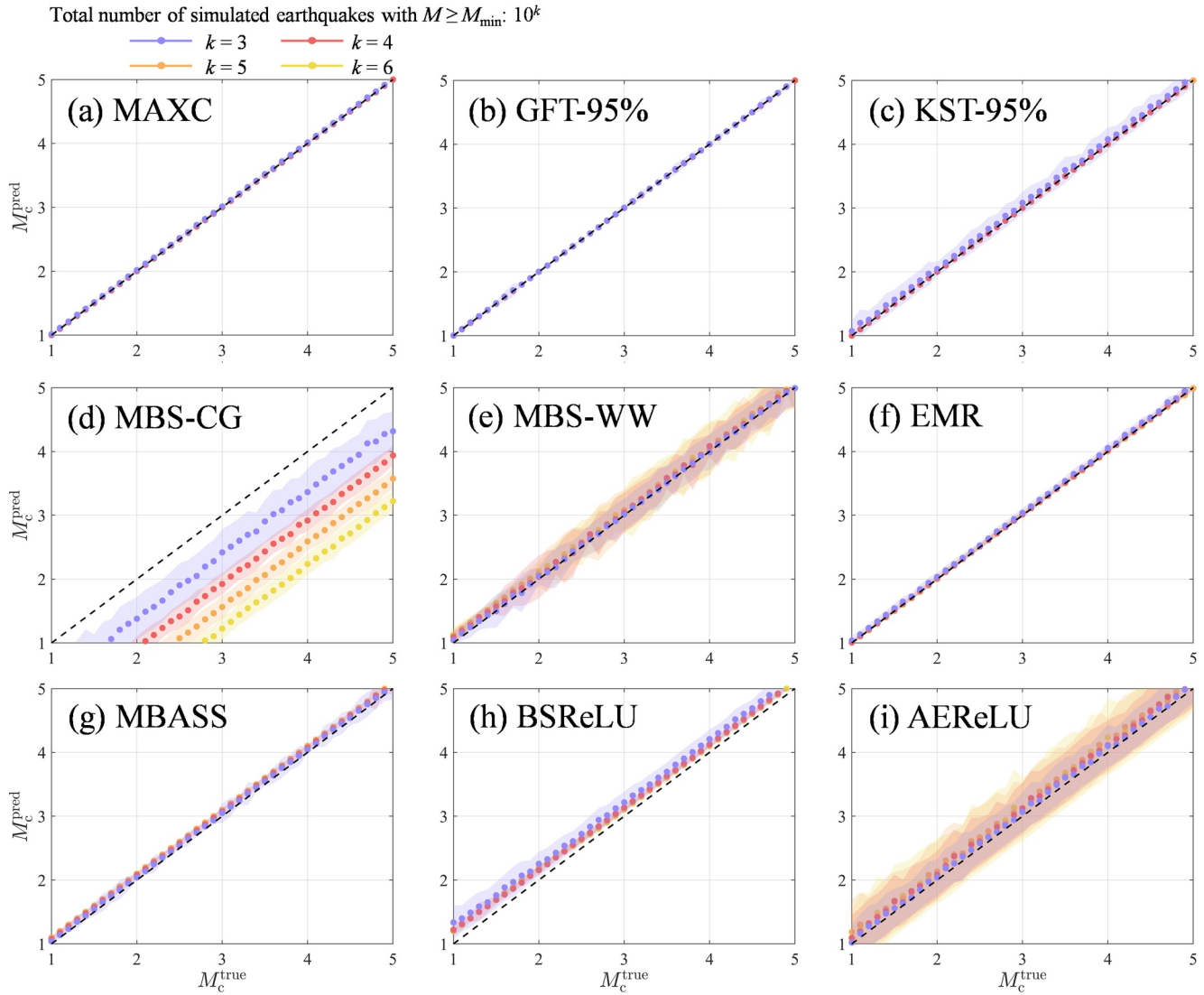


Figure 4. M_c^{pred} as a function of M_c^{true} for the simulated data sets with homogeneous M_c , using the AN (Angular) detection model (Section 3.1.2). Details are identical to those in Figure 3.

noticeably larger error bars, the uncertainties of the remaining estimates is minimal. This strong performance likely reflects the use of a single AN (Angular) detection model to generate the synthetic data, producing a sharp, well-defined break from the Gutenberg-Richter trend and thereby simplifying the estimation problem. By contrast, Figure 3 reveals, with the exception of EMR, which lacks M_c^{pred} values for M_c^{true} between 1.0 and 1.7 due to its minimum data requirement for fitting the incomplete portion ($M < M_c^{\text{true}}$), all methods provide reasonably accurate M_c^{pred} when M_c^{true} is small ($M_c^{\text{true}} \leq 2.0$). However, as M_c^{true} increases, all methods but BSReLU and AEReLU tend to underestimate M_c , regardless of the data set size. All methods, except for MBS-WW, BSReLU and AEReLU, fail to obtain reliable M_c^{pred} when M_c^{true} is larger than or equal to 3.5. This suggests that these methods may not be able to reliably and accurately estimate the short-term incompleteness of aftershock sequences following major earthquakes (e.g., Iwata, 2008).

In Figure 3, M_c^{pred} obtained by MAXC begins to deviate from M_c^{true} around $M_c^{\text{true}} = 2.0$, regardless of the data set size. This behavior is due to the method's sensitivity to the peak of the FMD. The performance of KST-95% is essentially consistent with that of GFT-95%, meaning that, if an estimation error of 0.2 is acceptable, reliable M_c^{pred} are obtained for $k = 3, 4, 5, 6$ at $M \leq M_c^{\text{true}} \approx 1.7, 3.0, 3.5, 3.5$, respectively (corresponding to the minimum number of events in the complete portion in Figure 2b, approximately 200, 100, 200, and 2,000). For MBS-CG, its

simplistic definition of b -value stability leads to the largest deviation of M_c^{pred} from M_c^{true} . The obtained M_c^{pred} values are only reliable for $k = 4, 5, 6$ when $M \leq M_c^{\text{true}} \approx 2.5$, and for $k = 3$ when $M \leq M_c^{\text{true}} \approx 2.0$. The drawbacks of EMR are very apparent, as it requires a sufficient amount of data in the incomplete region to fit the Gaussian distribution. This causes EMR to fail when M_c^{true} is very close to $M_{\min}$ and results in unstable estimates for small data sets ($k = 3$). For $k = 4, 5, 6$, EMR typically gives severely underestimated M_c^{pred} values when $M > M_c^{\text{true}} \approx 2.5$. MBS-WW appears to be a promising M_c estimation algorithm, as reliable results are obtained for $k = 3, 4, 5, 6$ at $M \leq M_c^{\text{true}} \approx 1.9, 2.7, 3.7, 4.5$, respectively, with the minimum number of events in the complete portion (Figure 2b) being approximately 100, 200, 200, and 300. This indicates that, when the number of events with $M \geq M_c$ in a catalog exceeds approximately 300, MBS-WW can provide very accurate and stable M_c^{pred} . MBASS exhibits a similar performance to MBS-WW, but the stability of the estimated M_c^{pred} is notably inferior to that of MBS-WW. In contrast, BSReLU and AEReLU are the only two methods that consistently produce M_c^{pred} values larger than M_c^{true} , regardless of the data set size. This slight overestimation largely reflects the flexibility of these new models in defining catalog completeness: recall that, in both BSReLU and AEReLU, M_c^{pred} is defined as the magnitude corresponding to a completeness probability of approximately 84%. Panels (h) and (i) in Figure 3 also clearly show that as the data set size decreases, the overestimation of M_c^{pred} becomes more pronounced. Overall, BSReLU tends to perform less favorably than AEReLU, highlighting the advantages of AEReLU's flexibility in allowing for an asymmetric transition. Remarkably, AEReLU is able to provide estimates close to M_c^{true} even when M_c^{true} is large, provided that the data size is sufficiently large. This suggests that AEReLU may be capable of reliably and accurately estimating the short-term incompleteness of aftershock sequences following major earthquakes. The results for synthetic catalogs obtained by iteratively pruning one, two, and four times show similar trends, with slight variations depending on the degree of data incompleteness controlled by the number of pruning operations.

4.2. Estimating M_c for Simulated Seismicity With Heterogeneous M_c

Figures 5 and 6 show the results of the estimation of M_c^{pred} as a function of the number of grids (2^N for $N = 1, 2, \dots, 10$) used to distribute the simulated earthquakes. Our seismicity distributing approach, which uses grids to simulate heterogeneity in M_c , essentially captures both the spatial and/or temporal variability of seismicity's M_c . Overall, except for MBS-WW, BSReLU and AEReLU, the other six methods tend to yield very small M_c^{pred} values when dealing with catalogs with heterogeneous M_c distributions (Figures 5 and 6). These estimates are often far below the mean M_c^{true} over all grids. Recall that our construction of the synthetic catalogs ensures that the mean M_c^{true} over all grids is equal to 3.0, with half of the grids having $M_c^{\text{true}} \leq 3.0$, and the other half having $M_c^{\text{true}} > 3.0$. Only MBS-WW, BSReLU and AEReLU, when the data set reaches a sufficient size (such as $k = 5$ and 6) and for higher-resolution grid scenarios (such as 2^N for $N \geq 3$), retrieves M_c^{pred} values that are close to the mean M_c^{true} across all grids (i.e., 3.0). For all methods except MBS-WW, BSReLU and AEReLU, M_c^{pred} , regardless of N , remain consistently between approximately 1.0 and 2.0.

4.3. Estimating M_c for Empirical Seismicity

For the six study regions with unknown M_c^{true} and spatiotemporally heterogeneous M_c in the observed earthquake catalogs, we used the nine catalog-based methods described above to estimate the mean M_c^{pred} and its uncertainty for each region, derived from 200 bootstrapping iterations (Efron, 1979). Table 1 reports the M_c estimates and their associated uncertainties for the seven methods based on the “cut-off M_c ” concept. In contrast, Table 2 presents results for BSReLU, additionally listing σ_{M_c} with its uncertainty, and for AEReLU, which includes both σ_{M_c} and the asymmetry parameter β , along with their respective uncertainties.

For the four study regions in the Chinese mainland (Beijing-Tianjin-Hebei region, the southeastern coastal region, the Sichuan-Yunnan region, and northern Xinjiang) and the two additional regions, California and New Zealand, the lowest mean M_c^{pred} is given by MAXC (Beijing-Tianjin-Hebei and southeastern coastal), and MBS-CG (Sichuan-Yunnan, northern Xinjiang, California, and New Zealand). Conversely, the largest mean M_c^{pred} values are consistently given by MBS-WW. For Beijing-Tianjin-Hebei, southeastern coastal, Sichuan-Yunnan, northern Xinjiang, and California, the mean M_c^{pred} values from MBS-WW are 1.75, 1.44, 1.71, 1.87, and 1.84, respectively—these values are the closest to 1.8 among all the methods. This value is noteworthy because the spatial boundaries of these regions were delineated using the $M_c \approx 1.8$ contour lines derived from the posterior M_c map generated by the BMC method in Li et al. (2023), which applied Bayesian corrections to M_c estimates using

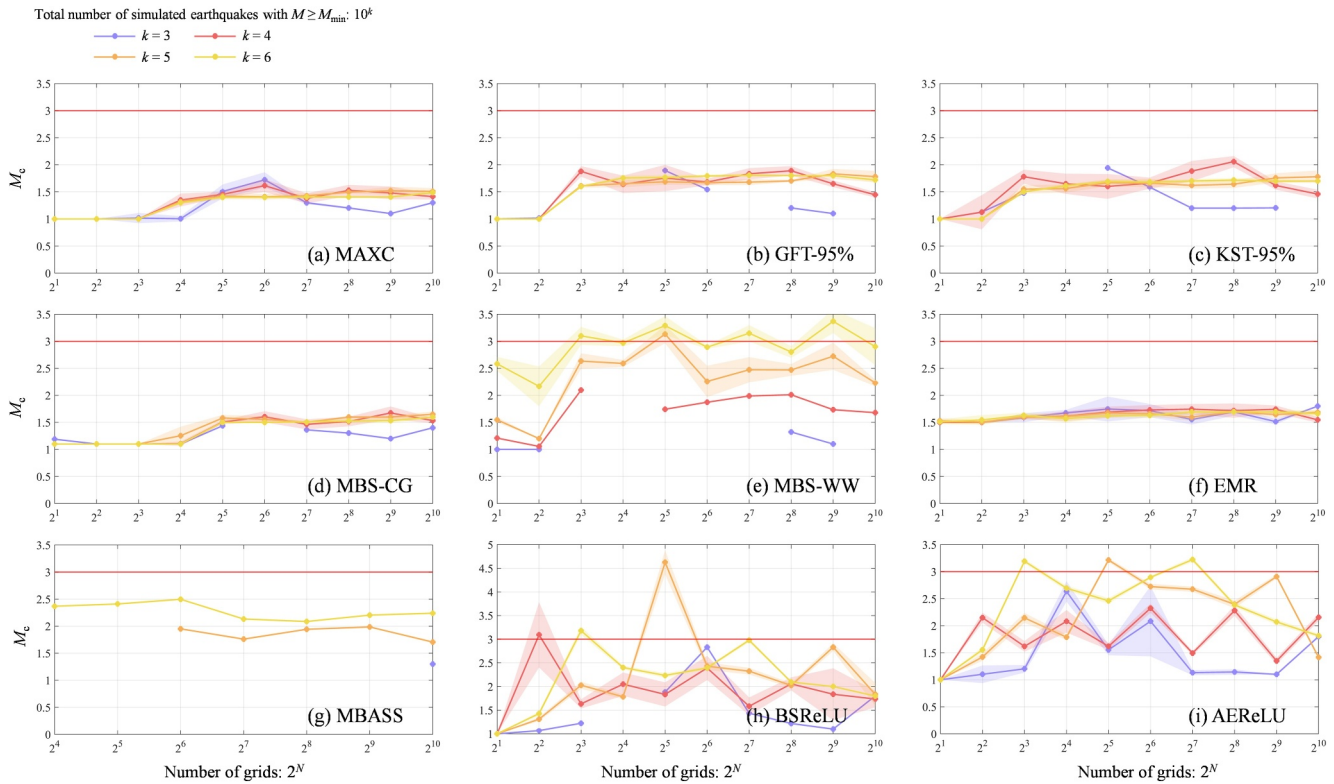


Figure 5. Results of the estimation of the completeness magnitude M_c as a function of the number of grids used to distribute the simulated earthquakes, after being iteratively pruned three times (Section 3.1.1). The total number of simulated earthquakes with $M \geq M_{\min}$, given by 10^k for $k = 3, 4, 5, 6$, is randomly and uniformly distributed across 2^N ($N = 1, 2, \dots, 10$) grids. The minimum completeness magnitude ($M_c^{\min} = 1.0$) and maximum completeness magnitude ($M_c^{\max} = 5.0$) are assigned to the first and last grids, respectively, with the remaining grids having completeness magnitudes linearly interpolated between these values. For example, with $2^N = 2^2$ grids, the completeness magnitudes are $M_c^{\text{grid}1} = 1.0$, $M_c^{\text{grid}2} = 2.3$, $M_c^{\text{grid}3} = 3.7$, and $M_c^{\text{grid}4} = 5.0$, ensuring that the average M_c across all grids is 3.0 (indicated by the red horizontal line in the figure). The shaded areas represent the uncertainties in the estimated completeness magnitude. The different colors correspond to the total number of simulated earthquakes. For details and methods used, refer to Figure 3.

prior information about network density. For New Zealand, the largest mean M_c^{pred} , 3.13, is also provided by MBS-WW.

However, the standard deviation of M_c^{pred} estimated by MBS-WW is relatively large compared to the other methods across the six study regions. The standard deviations for MBS-WW in Beijing-Tianjin-Hebei, south-eastern coastal, Sichuan-Yunnan, northern Xinjiang, California, and New Zealand are 0.17 (second largest), 0.20 (second largest), 0.19 (largest), 0.32 (largest), 0.10 (largest), and 0.35 (largest), respectively. For the six study regions, the method with the smallest M_c^{pred} standard deviation is KST-95% for Beijing-Tianjin-Hebei, EMR for southeastern coastal, MAXC for Sichuan-Yunnan, MBS-CG for northern Xinjiang, MAXC for California, and GFT-95% for New Zealand. Conversely, the method with the largest M_c^{pred} standard deviation is MBASS for Beijing-Tianjin-Hebei, southeastern coastal, and California, MBS-WW for Sichuan-Yunnan, northern Xinjiang, California, and New Zealand.

Table 2 presents more detailed results for the BSReLU and AEReLU models. Based on the simulation tests in Sections 4.1 and 4.2, we find the estimates from AEReLU to be more reliable. This is further supported by the significantly smaller uncertainty associated with the AEReLU parameter estimates compared to those of BSReLU, as shown in Table 2. Regarding AEReLU, the value M_c^{pred} defined at a completeness probability of approximately 84% indicates that the Beijing-Tianjin-Hebei region and the southeastern coastal region exhibit the highest data quality, with M_c^{pred} values below 2.0. These are followed by the Sichuan-Yunnan region and California, where M_c^{pred} values lie between 2.0 and 2.5. Notably, among all methods, AEReLU produces M_c^{pred} estimates closest to 1.8 in these four regions. This is consistent with the spatial boundaries of these regions, which were delineated using the $M_c \approx 1.8$ contour lines derived from the posterior M_c map of Li et al. (2023). In contrast, northern Xinjiang and New Zealand exhibit the poorest catalog completeness. According to AEReLU estimates in

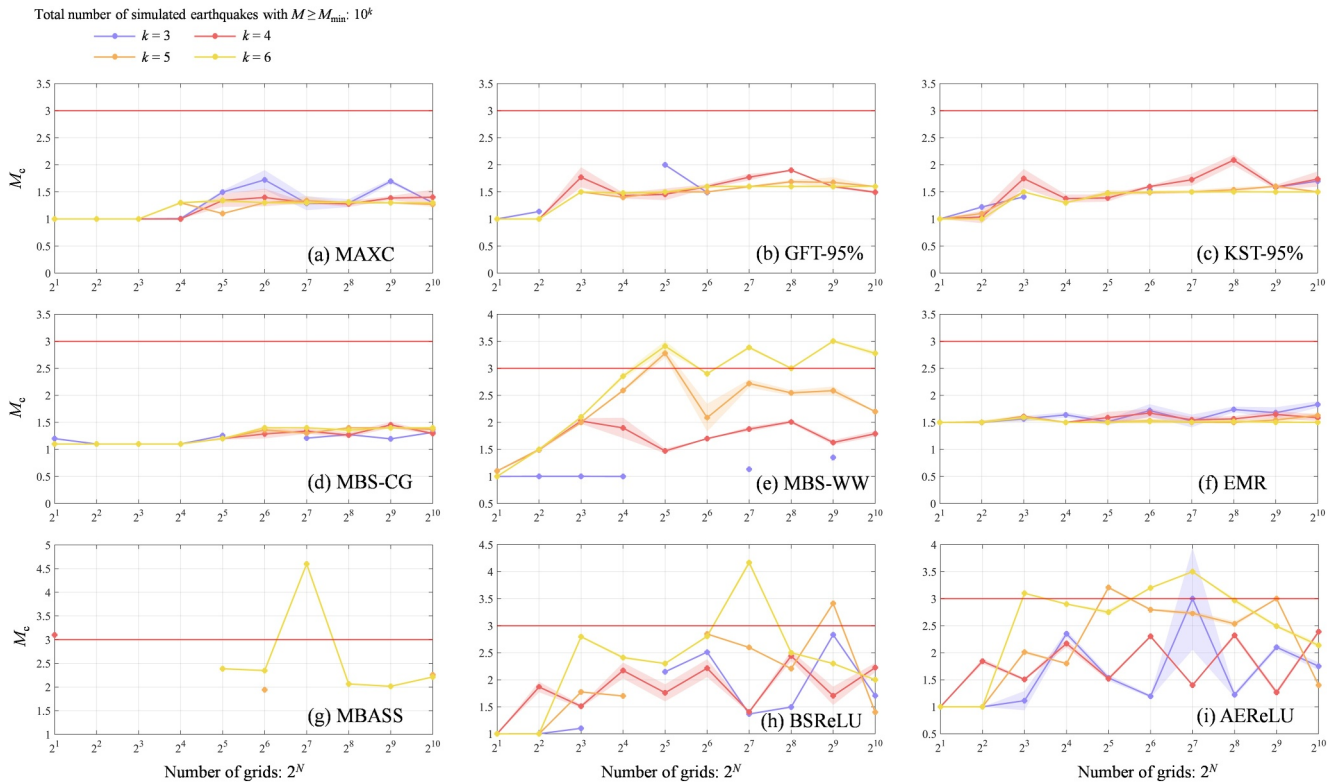


Figure 6. M_c as a function of the number of grids used to distribute the simulated earthquakes, using the AN (Angular) detection model (Section 3.1.2). Details are identical to those in Figure 5.

Table 2, even earthquakes with magnitudes above 4.0 in these regions have more than an 80% probability of being missing from the catalog. These results are consistent with the sparser seismic station coverage in northern Xinjiang and the higher proportion of deep-focus earthquakes in New Zealand.

The AEReLU results also show that, among the six study regions, Sichuan-Yunnan and northern Xinjiang yield mean b -values below 1.0, while the southeastern coastal region and Beijing-Tianjin-Hebei yield values closest to 1.0. In contrast, California and New Zealand exhibit mean b -values greater than 1.0. All six regions show relatively small standard deviations in b -value estimates, indicating that AEReLU provides precise and reliable b -value estimates across these regions. The estimated asymmetry parameters β for AEReLU are all smaller than 1 across the six regions (Table 2), indicating regional variability in detection completeness and asymmetric transitions around M_c . Specifically, $\beta < 1$ implies a slower convergence to the linear Gutenberg-Richter law in the $M \geq M_c$ domain, compared to the suppression of earthquakes below M_c . Beijing-Tianjin-Hebei, the southeastern coast, and California exhibit relatively larger β values (~ 0.5) with small uncertainties, suggesting a more gradual transition above M_c and contributing to the robustness of parameter estimation. In contrast, Sichuan-Yunnan

Table 1

Estimation Results of the Completeness Magnitude M_c and Their Associated Uncertainties Derived From 200 Bootstrapping Iterations, Using Seven Methods Based on the “Cut-Off M_c ” Concept Across the Six Study Regions

Region	MAXC	GFT-95%	KST-95%	MBS-CG	MBS-WW	EMR	MBASS
Beijing-Tianjin-Hebei	0.49 ± 0.10	1.01 ± 0.03	1.00 ± 0.01	0.86 ± 0.16	1.75 ± 0.17	0.80 ± 0.08	1.32 ± 0.64
Southeastern coastal	0.26 ± 0.06	0.92 ± 0.05	0.88 ± 0.05	0.34 ± 0.06	1.44 ± 0.20	0.60 ± 0.01	1.35 ± 0.63
Sichuan-Yunnan	1.10 ± 0.00	1.30 ± 0.01	1.30 ± 0.01	0.20 ± 0.00	1.71 ± 0.19	1.30 ± 0.00	1.60 ± 0.08
Northern Xinjiang	1.36 ± 0.06	1.59 ± 0.03	1.60 ± 0.05	0.20 ± 0.00	1.87 ± 0.32	1.57 ± 0.08	1.70 ± 0.07
California	0.90 ± 0.00	0.91 ± 0.00	0.91 ± 0.06	0.70 ± 0.00	1.84 ± 0.10	0.96 ± 0.10	0.91 ± 0.09
New Zealand	1.94 ± 0.05	2.17 ± 0.00	2.10 ± 0.00	0.20 ± 0.00	3.13 ± 0.35	2.00 ± 0.05	2.17 ± 0.05

Table 2

Estimation Results of the Parameters of BSReLU and AEReLU, Along With Their Associated Uncertainties Derived From 200 Bootstrapping Iterations Across the Six Study Regions

Region	BSReLU					AEReLU					
	a	b	M_c	σ_{M_c}	M_c^{pred}	a	b	M_c	σ_{M_c}	β	M_c^{pred}
Beijing-Tianjin-Hebei	5.85 ± 0.42	1.03 ± 0.07	0.62 ± 0.43	0.73 ± 0.21	1.97 ± 1.69	5.56 ± 0.12	0.97 ± 0.04	0.76 ± 0.09	0.38 ± 0.04	0.56 ± 0.11	1.65 ± 0.47
Southeastern coastal	5.62 ± 0.78	1.05 ± 0.11	0.74 ± 0.62	0.93 ± 0.28	2.03 ± 1.97	5.06 ± 0.13	0.96 ± 0.05	0.58 ± 0.10	0.32 ± 0.06	0.51 ± 0.10	1.35 ± 0.70
Sichuan-Yunnan	6.79 ± 0.02	0.92 ± 0.02	1.10 ± 0.01	0.95 ± 0.07	3.84 ± 0.53	6.72 ± 0.14	0.87 ± 0.01	1.09 ± 0.16	0.39 ± 0.02	0.39 ± 0.72	2.23 ± 0.22
Northern Xinjiang	6.08 ± 0.09	0.95 ± 0.03	1.36 ± 0.07	0.62 ± 0.09	2.83 ± 0.63	6.14 ± 0.14	0.92 ± 0.06	1.47 ± 0.08	0.39 ± 0.03	0.11 ± 0.20	4.08 ± 0.47
California	6.94 ± 0.38	1.23 ± 0.08	1.10 ± 0.31	0.88 ± 0.17	3.94 ± 2.23	6.55 ± 0.14	1.17 ± 0.04	0.96 ± 0.09	0.41 ± 0.05	0.46 ± 0.07	2.37 ± 0.47
New Zealand	7.45 ± 0.09	1.09 ± 0.02	1.95 ± 0.05	0.37 ± 0.04	2.91 ± 0.29	7.63 ± 0.10	1.11 ± 0.02	2.07 ± 0.06	0.39 ± 0.01	0.24 ± 0.16	4.06 ± 0.22

Note. Recall that, in both BSReLU and AEReLU, M_c^{pred} is defined as the magnitude corresponding to a completeness probability of approximately 84%.

(~ 0.4) shows a moderate β with large uncertainty, likely due to fewer data points in the $M \geq M_c$ range. New Zealand (~ 0.25) and northern Xinjiang (~ 0.10) yield the lowest β values, suggesting particularly gradual transitions to completeness in these regions.

5. Discussion: From Pointwise M_c to Distributed M_c

Within our evaluation framework, BSReLU and AEReLU deliver an impressive performance, and AEReLU demonstrates stronger stability in its estimations of M_c^{pred} . It is important to highlight, however, that the evaluation framework used in this study is based on the assumption that the “cut-off M_c ” concept is valid and fundamental. Specifically, this assumption posits that, above a certain M_c value, data is 100% complete, while below this M_c value, data is incomplete. Based on this assumption, we used the pruning function and the AN (Angular) detection model to generate data sets with a cut-off M_c . In contrast, the concept of M_c in BSReLU and AEReLU is represented as a probability density function, specifically a Gaussian distribution defined by a mean M_c and a standard deviation σ_{M_c} . Therefore, our evaluation framework, which assumes a cut-off M_c , might be inherently unfair to BSReLU and AEReLU. In other words, the testing framework represents a mis-specification for BSReLU and AEReLU, while it is a correct specification for the other seven methods, as they also operate on the basis of the cut-off M_c concept. When viewed in this context, it is particularly interesting to reassess the performance of BSReLU and AEReLU in this study (e.g., Li et al., 2025a).

Although EMR also incorporates a Gaussian complementary cumulative distribution function, it models the event detection probability of the seismic network rather than the completeness magnitude (Woessner & Wiemer, 2005). Specifically, EMR assumes that the probability of event detection by the network approaches 1 as M is increased, and approaches 0 as M is decreased. However, because the EMR framework cannot fully integrate the incomplete portion of FMD with the complete portion governed by the Gutenberg-Richter relationship, it still relies on the concept of a cut-off M_c . This assumes a step function where the event detection probability becomes exactly 1 for $M \geq M_c$, using Equation 1 to describe the CCFMD. In contrast, the BSReLU and AEReLU models overcome this limitation. They simultaneously capture the Gutenberg-Richter behavior of the logarithm of the CCFMD, which asymptotically aligns with a linear trend for $M \geq M_c$, while also incorporating a probabilistic representation of the completeness magnitude. This allows the BSReLU and AEReLU models to account for the fact that, even for magnitudes larger than M_c , there is a small but nonzero probability that some large earthquakes might go undetected. Similarly, for magnitudes smaller than M_c , the probability of detecting smaller earthquakes is not zero but progressively diminishes as the magnitude decreases.

Table 2 also shows that the σ_{M_c} values obtained with BSReLU vary widely across regions, indicating an uneven transition from incomplete to complete detection. Except for New Zealand, all regions have $\sigma_{M_c} > 0.6$; in the southeastern coastal, Sichuan-Yunnan, and California catalogs the estimates even exceed 0.9. By contrast,

AEReLU, which allows for an asymmetric transition produced by two distinct exponential decays, yields far more consistent widths, clustering around 0.3 to 0.4 for all six regions. The asymmetry parameter β reinforces this picture: Beijing-Tianjin-Hebei, the southeastern coastal zone, California, and Sichuan-Yunnan display relatively large values (0.4–0.5), New Zealand is intermediate (0.24), and northern Xinjiang is the smallest (0.11). In every case $\beta < 1$, implying that convergence to the linear Gutenberg-Richter branch for $M > M_c^{\text{pred}}$ is slower than the suppression of earthquakes below M_c^{pred} . This result challenges the common assumption of a sharp, fixed completeness magnitude and instead points to a gradual transition in which even some larger events remain unrecorded.

By integrating these features, the BSReLU and AEReLU models provide a more physically grounded framework for capturing the complexities of real-world earthquake catalogs, particularly those with incomplete entries. They offer greater flexibility and comprehensiveness in modeling compared to traditional approaches. Moreover, even for those seeking a straightforward and practical cut-off M_c value, the BSReLU and AEReLU models can provide useful recommendations. This is possible because one can back-calculate the corresponding magnitude threshold for a desired completeness probability, defined as the ratio between Equation 9, which characterizes the observed distribution, and Equation 1, which represents the ideal complete distribution. These models hold significant potential for future applications, particularly in statistical seismology frameworks that require simultaneous consideration of the Gutenberg-Richter law and magnitude incompleteness. For example, it could be highly beneficial in models like the Epidemic-Type Aftershock Sequence (ETAS) model (Hawkes & Oakes, 1974; Kagan, 1991; Li & Sornette, 2025; Li, Sornette, Wu, et al., 2025; Ogata, 1988, 1998; Ogata & Zhuang, 2006; Sornette & Werner, 2005), which heavily depend on accurate representations of both complete and incomplete seismicity.

6. Conclusions

Analyzing the magnitude of completeness (M_c) is an indispensable first step in nearly all earthquake statistical analyses. In this process, catalog-based methods for estimating M_c are the simplest, most straightforward, and most commonly used approach. Moreover, they often serve as the foundation for more advanced M_c estimation techniques. Therefore, it is essential to identify which methods are most reliable and to determine the conditions under which they perform best. In this study, we evaluated and compared nine catalog-based methods, including three newly developed methods for estimating M_c , which leverage our expertise in statistical testing and computer science. These methods were systematically analyzed within a robust framework designed to evaluate their performance under varying conditions, including data sets with homogeneous and heterogeneous M_c distributions, as well as real-world earthquake catalogs from regions such as China, California, and New Zealand.

For simulated earthquake catalogs with homogeneous M_c , all methods provide reasonably accurate M_c^{pred} when M_c^{true} is very low ($M_c^{\text{true}} \leq 2.0$). However, when M_c^{true} is greater than or equal to 3.5, six of the methods fail to produce reliable estimates, with the exception of MBS-WW, BSReLU and AEReLU. MBS-WW is the only method based on “the cut-off M_c ” concept capable of generating reliable results under the condition that the minimum number of events in the complete portion is approximately 300 or more. BSReLU and AEReLU are the only two methods that consistently produce M_c^{pred} values larger than M_c^{true} , regardless of the data set size. Remarkably, AEReLU is able to provide estimates close to M_c^{true} even when M_c^{true} is large, provided that the data size is sufficiently large. For catalogs with heterogeneous M_c distributions, the other six methods tend to yield very small M_c^{pred} values, which are often significantly below the mean M_c^{true} across all grids. In contrast, MBS-WW, BSReLU and AEReLU achieve M_c^{pred} estimates close to the mean M_c^{true} (3.0) across all grids when data sets are sufficiently large (e.g., for 10^5 or 10^6 simulated earthquakes) and grid resolutions are high (e.g., with more than 2^3 grid elements). BSReLU tends to perform less favorably than AEReLU, highlighting the advantages of AEReLU's flexibility in allowing for an asymmetric transition.

For the empirical earthquake catalogs in the six study regions, MBS-WW consistently yields mean M_c^{pred} values close to $M_c \approx 1.8$, in agreement with the spatial boundaries defined by posterior M_c maps (Li et al., 2023). However, its estimates exhibit relatively large uncertainties (standard deviations ranging from 0.10 to 0.35), and it fails to provide reliable M_c values in northern Xinjiang and New Zealand—likely due to sparse station coverage and a high proportion of deep-focus earthquakes, respectively. In contrast, AEReLU not only produces M_c^{pred} values closest to 1.8 in Beijing-Tianjin-Hebei, the southeastern coastal region, Sichuan-Yunnan, and California, but also offers a more flexible framework by defining M_c^{pred} at a completeness probability of $\sim 84\%$. This reflects

the highest catalog completeness in the first two regions ($M_c^{\text{pred}} < 2.0$), followed by Sichuan-Yunnan and California ($M_c^{\text{pred}} \approx 2.0\text{--}2.5$), with the poorest completeness in northern Xinjiang and New Zealand, where even events with $M \geq 4.0$ may have a larger than 80% chance of being undetected. Moreover, AEReLU provides precise and consistent b -value estimates across all regions, with $b < 1$ in Sichuan-Yunnan and northern Xinjiang, $b \approx 1$ in the southeastern coast and Beijing-Tianjin-Hebei, and $b > 1$ in California and New Zealand. The estimated asymmetry parameter β is below 1 in all regions, indicating that completeness transitions are asymmetric and typically more gradual above M_c^{pred} . Notably, regions with more gradual transitions (e.g., Beijing-Tianjin-Hebei, southeastern coast, California) exhibit higher β values (~ 0.5) and lower uncertainty, whereas northern Xinjiang and New Zealand display the lowest β values ($\sim 0.10\text{--}0.25$), reflecting particularly slow transitions to catalog completeness.

The above results have been derived by following the tradition of considering that the transition from the incomplete part to the complete part of catalogs is sharp, occurring at a well-defined so-called “magnitude of completeness” M_c . This is a convenient assumption made for the sake of simplicity but it is unlikely to be correct. Indeed, rather than being abrupt, completeness appears to unfold gradually, as a significant number of earthquakes with magnitudes exceeding M_c , and thus theoretically detectable, still go unrecorded in the catalogs. This is the motivation for our introduction of two new models, BSReLU and AEReLU, which represent the technically and physically realistic progressive transition from the “complete” regime where all earthquakes are measured to the “incomplete” part of the catalog where events are less and less likely to be detected, the smaller their magnitudes. Unlike traditional methods that rely on a strict cut-off M_c , BSReLU and AEReLU address the issue of incompleteness as a probabilistic concept. BSReLU and AEReLU capture the gradual transition between incomplete and complete data, reflecting the inherent uncertainties in event detection probabilities. BSReLU models the likelihood of an earthquake being observed using a Gaussian distribution, defined by a mean M_c and a standard deviation σ_{M_c} . AEReLU allows for an asymmetric transition between the incomplete part to the complete part of the catalog, governed by two potentially different rates of convergence (rate equal to 1 for $M < M_c$ and rate equal to β for $M \geq M_c$). Applied to empirical catalogs, AEReLU revealed that the convergence to the linear Gutenberg-Richter branch for $M > M_c^{\text{pred}}$ is slower than the suppression of earthquakes below M_c^{pred} . This suggests that the concept of completeness is more gradual than traditionally modeled, with a non-negligible fraction of earthquakes remaining undetected and absent from the catalogs despite having magnitudes above M_c and being theoretically detectable.

Our evaluation framework based on synthetic catalogs has assumed a fixed well-defined cut-off M_c . In this context, MBS-WW, BSReLU and AEReLU overall have been found to be the best methods to determine the magnitude of completeness M_c . We should stress that this set-up is likely to inherently disadvantage BSReLU and AEReLU. In other words, for these two methods, the synthetic tests represent a case of model mis-specification, as they do not reflect their underlying assumptions. In contrast, the tests are well aligned with the assumptions of the other seven methods, which also rely on the cut-off M_c concept. From this perspective, it becomes especially important to reevaluate the performance of BSReLU and AEReLU under a different framework in which M_c varies smoothly rather than being fixed, as assumed in the present study (e.g., Li et al., 2025a).

BSReLU and AEReLU offer significant advantages, especially in cases where the strict cut-off assumption oversimplifies the complexities of real-world seismicity. For instance, for magnitudes larger than M_c , BSReLU and AEReLU account for the possibility of missing events, while for smaller magnitudes, they effectively model the decreasing probability of detection as the magnitude diminishes. This probabilistic framework provides a more realistic representation of seismicity, particularly for catalogs with varying levels of completeness. Unlike traditional methods, including MBS-WW, which assume that all data above a certain M_c are 100% complete, BSReLU and AEReLU challenge this assumption by offering a more flexible and probabilistic perspective on completeness. This flexibility enables BSReLU and AEReLU to bridge the gap between traditional cut-off models and probabilistic frameworks, making them particularly well-suited for applications requiring a detailed understanding of data completeness. Note that BSReLU and AEReLU are just two implementations of the concept that the transition from complete to incomplete ought to be described by a gradual probabilistic model. Other parameterizations are possible, for instance by replacing the cumulative Gaussian function by other sigmoid functions with different left and right tails. Given these promising attributes, we believe that continued attention and exploration of this class of models are warranted for future research and applications. Moreover, novel testing methods need to be developed that are adapted to their gradual probabilistic structure.

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Data Availability Statement

The California catalog is sourced from the Advanced National Seismic System (ANSS) Comprehensive Earthquake Catalog (ComCat; U.S. Geological Survey, 2017). For the catalog in New Zealand, the data are obtained from the GeoNet Earthquake Catalog of New Zealand (GNS Science, 2022). We acknowledge the New Zealand GeoNet programme and its sponsors EQC, GNS Science, LINZ, NEMA and MBIE for providing data used in the present study. The data for the four study regions in China are acquired from the China Earthquake Networks Center (CENC) through the internal link provided by the Earthquake Cataloging System at China Earthquake Administration. The earthquake catalog for the four study regions in China, covering events with magnitudes ≥ 0 from 2009 to 2024 (excluding time and location information), along with the MATLAB codes for BSReLU and AEReLU developed in this study, are openly available at Zenodo (Li et al., 2025b).

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